



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010

YEAR 12

ASSESSMENT TASK #3

Mathematics Extension 1

General Instructions

- Working time – 90 Minutes
- Reading Time – 5 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each **Section** is to be returned in a separate booklet.
- All necessary working should be shown in every question if full marks are to be awarded.
- Full marks may not be awarded for untidy or badly arranged work.

Total Marks – 72

- Attempt questions 1 – 6
- All questions are NOT of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *A.M.Gainford*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A
(Start a new booklet.)

Question 1. (12 marks)

Marks

- (a) Prove the following identity: **2**

$$\frac{\sin A}{\cos A + \sin A} + \frac{\sin A}{\cos A - \sin A} = \tan 2A$$

- (b) Find: $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ **1**

- (c) Use the substitution $u = 2x - 1$ to find **4**

$$\int_{\frac{1}{2}}^1 4x(2x-1)^3 dx$$

- (d) The sector AOB of a circle centre O and radius r has an area of $\frac{3\pi}{4} \text{ cm}^2$. If the arc AB subtends an angle of $\frac{\pi}{6}$ at O , find the length of the arc AB . **2**

- (e) Give the solutions of the equation $\sin\left(\theta - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ in $0 \leq \theta \leq 2\pi$. **3**

Question 2. (12 Marks)

- (a) Differentiate the following with respect to x : **6**

(i) $x^2 \tan 2x$

(ii) $\cos^{-1} 3x$

(iii) $\log_e(\cos x)$

- (b) Find an indefinite integral of each of the following (with respect to x): **6**

(i) $\frac{1}{\sqrt{4-x^2}}$

(ii) $\sin(1-3x)$

(iii) $\sin x \cos^3 x$

Section B
(Start a new booklet.)

Question 3. (12 marks)

- (a) (i) State the domain and range of the function $f(x) = 3\sin^{-1}\left(\frac{x}{2} - 1\right)$. 5
- (ii) Sketch the graph of $y = f(x)$.
- (iii) Evaluate $f(1)$.
- (b) Use the substitution $u = \cot x$ to evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^2 x \operatorname{cosec}^2 x \, dx$. 3
- (c) The equation $x^3 - 103x + 102 \cdot 5 = 0$ has a root near $x = 1$. 2
Take $x = 1$ as a first approximation and use Newton's method once to obtain a closer approximation to this root.
- (d) A particle moves along a straight line so that its distance from a fixed point O at a time t is given by $x = 2 \cos\left(\frac{t}{4}\right)$. 2
What is the exact value of the acceleration of the particle when $t = \frac{4\pi}{3}$?

Section continues overleaf

Question 4. (12 marks)

Marks

- (a) A particle P moves along the x -axis so that at time t seconds it is x cm from the origin O and its velocity is v cm/s. Initially the particle is at rest at the origin. **6**
- (i) If the acceleration of P is given by $\ddot{x} = 4(40 - x)$ cm/s², use $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ to show $v^2 = 4(80x - x^2)$.
- (ii) Prove that P moves in the interval $0 \leq x \leq 80$.
- (iii) Find the maximum velocity of the particle, and where the maximum occurs.
- (b) The inside of a vessel used for water has the shape of a solid of revolution obtained by the rotation of the parabola $9y = 8x^2$ about the y -axis. **6**
The depth of the vessel is 8 cm.
- (i) Prove that the volume of water, when the depth is h cm, is $\frac{9}{16} \pi h^2$ cm³.
- (ii) If water is poured into the vessel at a rate of 20 cm³/s, find the rate at which the water is rising when the depth is 4 cm.

Section C
(Start a new booklet.)

Question 5. (12 marks)

- | | Marks |
|---|--------------|
| (a) The digits 1, 2, 3, 4, 5 are arranged in a row. | 4 |
| <div style="margin-left: 40px;">(i) In how many ways can this be done without restriction?</div> <div style="margin-left: 40px;">(ii) In how many of these arrangements will there be two or more odd numbers together?</div> | |
| (b) Find the exact value of $\int_0^{\frac{\pi}{8}} \sin 2x \cos^2 2x dx$. | 2 |
| (c) Find $\cos \theta$ if $\theta = \cos^{-1} \frac{24}{25} - \sin^{-1} \frac{15}{17}$. | 2 |
| (d) A cylindrical hole of radius c is bored symmetrically through a sphere of radius a .
Show that the remaining volume is $\frac{4}{3}\pi(a^2 - c^2)^{\frac{3}{2}}$. | 4 |

Section continues overleaf

Question 6. (12 marks)

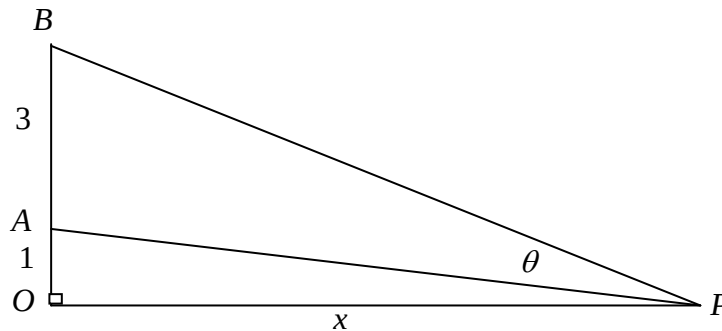
Marks
6

(a) Consider the functions $y = 1 + \cos \theta$ and $y = \sin^2 \theta$.

- (i) Sketch the graphs of the curves, on the same axes, in the domain $-\pi \leq \theta \leq \pi$.
- (ii) Show algebraically that the curves intersect at $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.
- (iii) Find the area enclosed by the curves between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.

(b)

6



In the diagram, a vertical pole AB , 3 metres high, is placed on top of a support 1 metre high. The pole subtends an angle of θ radians at the point P , which is x metres from the base O of the support.

- (i) Show that $\theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$.
- (ii) Show that θ is a maximum when $x = 2$.
- (iii) Deduce that the maximum angle subtended at P is $\theta = \tan^{-1} \frac{3}{4}$.

This is the end of the paper.

QUESTION ONE

$$\theta = \pi/6$$

$$\frac{1}{2}r^2\left(\frac{\pi}{6}\right) = 3\pi/4$$

$$\frac{1}{2}r^2 = 9/2$$

$$r^2 = 9$$

$$r = 3$$

$$l = r\theta$$

$$= 3 \times \pi/6$$

$$= \frac{\pi}{2}$$

(2)

$$a) \text{ LHS} = \frac{\sin A}{\cos A + \sin A} + \frac{\sin A}{\cos A - \sin A}$$

$$= \frac{\sin A(\cos A - \sin A) + \sin A(\cos A + \sin A)}{(\cos A + \sin A)(\cos A - \sin A)}$$

$$= \frac{\sin A \cos A - \sin^2 A + \sin A \cos A + \sin^2 A}{\cos^2 A - \sin^2 A}$$

$$= \frac{2\sin A \cos A}{\cos 2A}$$

$$= \frac{\sin 2A}{\cos 2A}$$

(2)

$$= \tan 2A$$

$$= \text{RHS}$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 3x}{3} = \lim_{x \rightarrow 0} 3 \frac{\sin 3x}{3}$$

$$= (3)(1)$$

$$= 3$$

(1)

$$c) \int_{\frac{1}{2}}^1 4x(2x-1)^3 dx$$

$$\text{Let } u = 2x - 1 \Rightarrow x = \frac{u+1}{2}$$

$$du = 2 \cdot dx$$

$$\text{when } x = \frac{1}{2} \quad u = 0$$

$$\text{when } x = 1 \quad u = 1$$

$$\int_{\frac{1}{2}}^1 4x(2x-1)^3 dx = \int_0^1 2 \left(\frac{u+1}{2} \right) (u^3) du$$

$$= \int_0^1 u^4 + u^3 du$$

$$= \left[\frac{1}{5} u^5 + \frac{1}{4} u^4 \right]_0^1$$

$$= \left(\frac{1}{5} + \frac{1}{4} \right) - 0$$

$$= \frac{9}{20}$$

(4)

$$e) \sin\left(\theta - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\sin\left(\theta - \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)$$

$$\theta - \pi/4 = n\pi + (-1)^n \frac{\pi}{4}$$

$$\theta = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}$$

$$= \frac{\pi}{4} (4n + (-1)^n + 1)$$

when n is odd

$$\theta = \pi n$$

when n is even

$$\theta = \pi/2 (2n+1)$$

$$\sin\left(\theta - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\sin\left(\theta - \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)$$

$$\theta - \pi/4 = \pi/4, 3\pi/4$$

$$\theta = \pi/2, \pi$$

in domain $0 < \theta < 2\pi$ (3)

QUESTION TWO.

a)(i) $\frac{d}{dx}(x^2 \tan 2x)$

$= uv' + u'v$

$u = x^2 \quad v = \tan 2x \quad (2)$

$u' = 2x \quad v' = 2 \sec^2 2x$

$= x^2(2 \sec^2 2x) + 2x(\tan 2x)$

$= 2x^2 \sec^2 2x + 2x \tan 2x$

$= 2x(x \sec^2 2x + \tan 2x)$

(ii) $\frac{d}{dx}(\cos^{-1} 3x)$

$= \frac{d}{dx}(\cos^{-1} \frac{x}{\frac{1}{3}}) = \frac{-1}{\sqrt{(\frac{1}{3})^2 - x^2}}$

$= \frac{-1}{\sqrt{1 - 9x^2}}$

$= \frac{-3}{\sqrt{1 - 9x^2}} \quad (2)$

(iii) $\frac{d}{dx}(\ln(\cos x))$

$= \frac{1}{\cos x} (-\sin x)$

$= -\tan x \quad (2)$

b)(i) $\int \frac{1}{\sqrt{4-x^2}} dx = \sin^{-1} \frac{x}{2} + C \quad (2)$

ii) $\int \sin(1-3x) dx \quad (2)$

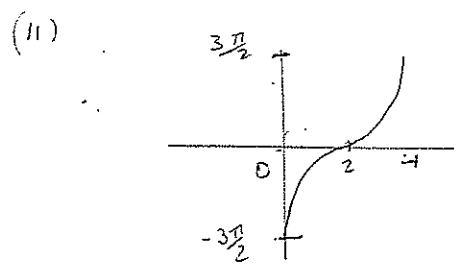
$= -\frac{1}{3} \cos(1-3x) + C$

$= \frac{1}{3} \cos(1-3x) + C$

ii) $\int \sin x \cos^3 x dx = \frac{-\cos^4 x}{4} + C \quad (2)$

QUESTION 3

(i) Domain: $0 \leq x \leq 4$
 Range: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$



(iii) $f(1) = 3 \sin^{-1}(-\frac{1}{2}) = -\frac{\pi}{2}$

(b) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^2 x \operatorname{cosec}^2 x \, dx$

$u = \cot x$
 $\frac{du}{dx} = -\operatorname{cosec}^2 x$

$-\int_0^1 \frac{u^2 \operatorname{cosec}^2 x \, du}{\operatorname{cosec}^2 x}$

$x = \frac{\pi}{2} \quad u = 0$
 $x = \frac{\pi}{4} \quad u = 1$

$\left[\frac{u^3}{3} \right]_1^0 = \frac{1}{3} - 0$
 $= \frac{1}{3}$

(c) $x^3 - 103x + 102.5 = 0$ root near 1

$a_1 = 1 - \frac{f(1)}{f'(1)} \quad f(x) = x^3 - 103x + 102.5$

$a_1 = 1 - \frac{0.5}{-100}$

$a_1 = 1.005$

(d) $x = 2 \cos\left(\frac{t}{4}\right)$

$v = \frac{dx}{dt} = -\frac{1}{2} \sin\left(\frac{t}{4}\right)$

$a = \frac{dv}{dt} = -\frac{1}{8} \cos\left(\frac{t}{4}\right)$

$t = \frac{4\pi}{3} \quad a = -\frac{1}{8} \cos \frac{\pi}{3}$

$a = -\frac{1}{16} \text{ m/s}^2$

QUESTION 4

(a) $x = 4(40 - x) \quad t=0 \quad x=0 \quad v=0$

(i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4(40 - x)$

$\frac{1}{2} v^2 = 160x - 2x^2 + C$

$v=0 \quad x=0 \quad C=0$

$v^2 = 320x - 4x^2 = 4(80x - x^2)$

(ii) from (i) $v=0$ when $4(80x - x^2) = 0$

$x = 0, 80$

particle moves between $x=0$ and $x=80$

(iii) Max velocity $\frac{dx}{dt} = 0$

$v = 2\sqrt{80 \times 40 - 40^2}$

$\text{max} = 80 \text{ m/s}$

4(b)(i) $ay = 2x^2$

$x^2 = \frac{a}{8} y$

$V = \pi \int_0^h \frac{a}{8} y \, dy$

$= \pi \left[\frac{a}{16} y^2 \right]_0^h$

$= \frac{a}{16} \pi h^2 \text{ cm}^3$

(ii) $\frac{dv}{dt} = 20 \quad \frac{dv}{dh} = \frac{a}{8} \pi h$ (from (i))

$\frac{dh}{dt} = \frac{dv}{dt} \times \frac{dh}{dv}$

$= 20 \times \frac{8}{9\pi \times 4}$

$= \frac{40}{9\pi} \text{ cm/s}$

$\approx 1.415 \text{ cm/s}$

SECTION C

Q5 (a)(i) $5! = \boxed{120}$

2

(ii) $\underline{0} \underline{\in} \underline{0} \underline{\in} \underline{0}$

This is the only possibility with all odd separated.

$5! - 2! \times 3! = \boxed{108}$

2

(b) $\int_0^{\frac{\pi}{8}} \sin 2x \cos^2 2x dx$

let $u = \cos 2x$

$\frac{du}{dx} = -2 \sin 2x$

$= -\frac{1}{2} \int_1^{\frac{1}{\sqrt{2}}} u^2 du$

$-\frac{1}{2} du = \sin 2x dx$

$= -\frac{1}{2} \left[\frac{u^3}{3} \right]_1^{\frac{1}{\sqrt{2}}}$

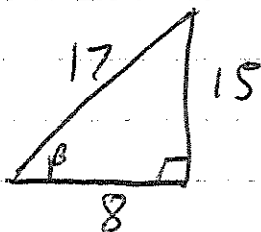
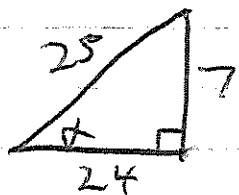
$= -\frac{1}{6} \left[u^3 \right]_1^{\frac{1}{\sqrt{2}}}$

$= -\frac{1}{6} \left(\left(\frac{1}{\sqrt{2}} \right)^3 - 1 \right)$

$= \boxed{\frac{1}{6} - \frac{1}{12\sqrt{2}}}$

2.

(c)



$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

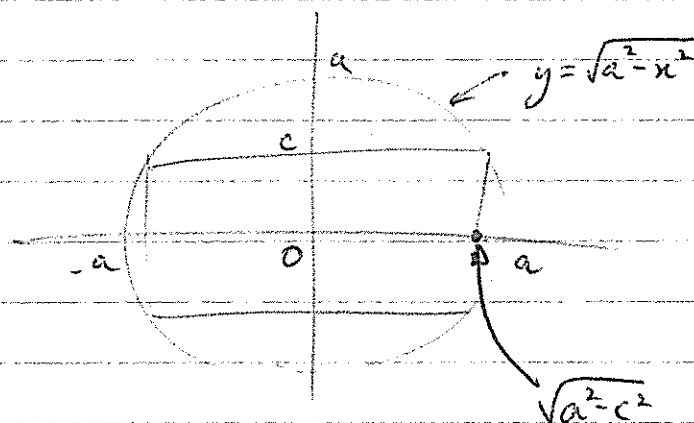
2

$= \frac{24}{25} \times \frac{8}{17} + \frac{7}{25} \times \frac{15}{17}$

$= \boxed{\frac{297}{425}}$

sphere radius a
cylindrical hole radius c .

$$\frac{4}{3}\pi(a^2 - c^2)^{3/2}$$



when $y = c$

$$c = \sqrt{a^2 - x^2}$$

$$c^2 = a^2 - x^2$$

$$x^2 = a^2 - c^2$$

$$x = \pm \sqrt{a^2 - c^2}$$

$$V = \pi \int_a^b y^2 dx$$

$$\frac{1}{2}V = \pi \int_0^{\sqrt{a^2 - c^2}} (a^2 - x^2) dx - \pi \int_0^{\sqrt{a^2 - c^2}} c^2 dx$$

$$= \pi \left[a^2 x - \frac{x^3}{3} \right]_0^{\sqrt{a^2 - c^2}} - \pi \left[c^2 x \right]_0^{\sqrt{a^2 - c^2}}$$

$$= \pi \left[a^2 \sqrt{a^2 - c^2} - \frac{(a^2 - c^2)\sqrt{a^2 - c^2}}{3} - 0 \right] - \pi \left[c^2 \sqrt{a^2 - c^2} - 0 \right]$$

$$= \pi \left[a^2 \sqrt{a^2 - c^2} - \frac{a^2 \sqrt{a^2 - c^2}}{3} + \frac{c^2 \sqrt{a^2 - c^2}}{3} - c^2 \sqrt{a^2 - c^2} \right]$$

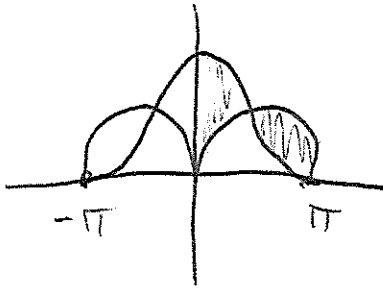
$$= \pi \left[\frac{2a^2 \sqrt{a^2 - c^2}}{3} - \frac{2c^2 \sqrt{a^2 - c^2}}{3} \right]$$

$$= \frac{2\pi}{3} \left[\sqrt{a^2 - c^2} (a^2 - c^2) \right]$$

$$= \frac{2\pi}{3} (a^2 - c^2)^{3/2}$$

$$V = \frac{4\pi}{3} (a^2 - c^2)^{3/2}$$

Q6 a)(i) $y = 1 + \cos \theta$ $y = \sin^2 \theta$.



2

(ii) $1 + \cos \theta = 1 - \cos^2 \theta$.

$$\cos^2 \theta + \cos \theta = 0.$$

$$\cos \theta (\cos \theta + 1) = 0.$$

~~(i)~~ $\cos \theta = 0$

$$\theta = \frac{\pi}{2}, -\frac{\pi}{2}$$

2

$$\cos \theta = -1$$

$$\theta = \pi, -\pi.$$

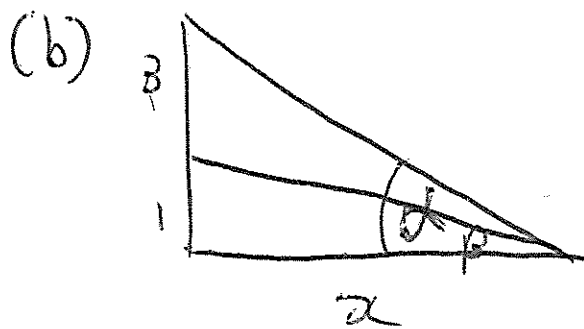
(ii) $2 \int_0^{\pi/2} 1 + \cos \theta - \sin^2 \theta d\theta + 2 \int_{\pi/2}^{\pi} \sin^2 \theta - 1 - \cos \theta d\theta$

$$= \int_0^{\pi/2} 2 + 2\cos \theta - 1 + \cos 2\theta d\theta + \int_{\pi/2}^{\pi} -1 - \cos 2\theta - 2\cos \theta d\theta$$

$$= \left[\theta + 2\sin \theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2} + \left[-\theta - \frac{1}{2} \sin 2\theta - 2\sin \theta \right]_{\pi/2}^{\pi}$$

$$= \left(\frac{\pi}{2} + 2 + 0 \right) - (0 + 0 + 0) + (-\pi - 0 - 0) - \left(-\frac{\pi}{2} - 0 - 2 \right)$$

$$= \frac{\pi}{2} + 2 - \pi + \frac{\pi}{2} + 2 = 4$$



$$\theta = \alpha - \beta.$$



(i) $\tan \alpha = \frac{4}{x} \quad \tan \beta = \frac{1}{x}.$

$$\theta = \tan^{-1} \frac{4}{x} - \tan^{-1} \frac{1}{x}$$

(ii). $\theta' = \frac{1}{1+(\frac{4}{x})^2} \times -\frac{4}{x^2} - \frac{1}{1+(\frac{1}{x})^2} \times -\frac{1}{x^2}.$

$$= \frac{1}{x^2+1} - \frac{4}{x^2+16}$$

$$\theta'(x) = \frac{12-3x^2}{(x^2+1)(x^2+16)}$$

$\theta'(x) = 0$ when $12-3x^2 = 0.$

$$x = \pm 2.$$

$$\theta'(1) = \frac{12-3}{(2)(17)} > 0 \quad \theta'(3) = \frac{12-27}{(10)(25)} < 0$$

$\therefore x=2$ is a maximum.

(iii) $\theta(2) = \tan^{-1}(2) - \tan^{-1}(\frac{1}{2}).$

$$\begin{aligned} \tan(\theta(2)) &= \tan(\tan^{-1} 2 - \tan^{-1} \frac{1}{2}). \\ &= \frac{2 - \frac{1}{2}}{1 + 1^2} \\ &= \frac{\frac{3}{2}}{\frac{3}{2}} = \frac{3}{4}. \end{aligned}$$

i.e. $\theta = \tan^{-1}(\frac{3}{4}).$